

Data Visualizations

RQ1: What Key Factors Lead to Late Deliveries, and How Can We Proactively Identify High-Risk Orders to Improve Customer Satisfaction?

This section outlines the performance evaluation of five classification models built to predict late deliveries: Logistic Regression, Decision Tree, Random Forest, XGBoost, and Artificial Neural Network (ANN). The main objective was to improve recall for the minority class (late deliveries), which comprises approximately 6–10% of the dataset.

Dataset Characteristics:

- Dataset Size after Cleaning: 108,983 Rows, 12 Columns
- Target Variable: delivered_late
- Input Features:
 - **Numerical:** handoff_delay, distance_km, product_weight_g, product_length_cm, product_height_cm, product_width_cm, product_volume
 - **Categorical:** customer_city, seller_city, seller_id, product_category_name_english
- Highly Imbalanced: ~90% on-time deliveries, ~10% late deliveries
- Resampling: **SMOTE** applied to training data

All models were evaluated on a separate test set using the following metrics:

- Recall (priority)
- F2 Score (weighted for recall)
- ROC-AUC
- Confusion Matrix

Recall was prioritized as late deliveries are rare, accuracy is misleading (a model could achieve ~90% by predicting all as on-time). Recall was prioritized to reduce false negatives—ensuring more actual late deliveries are correctly identified.

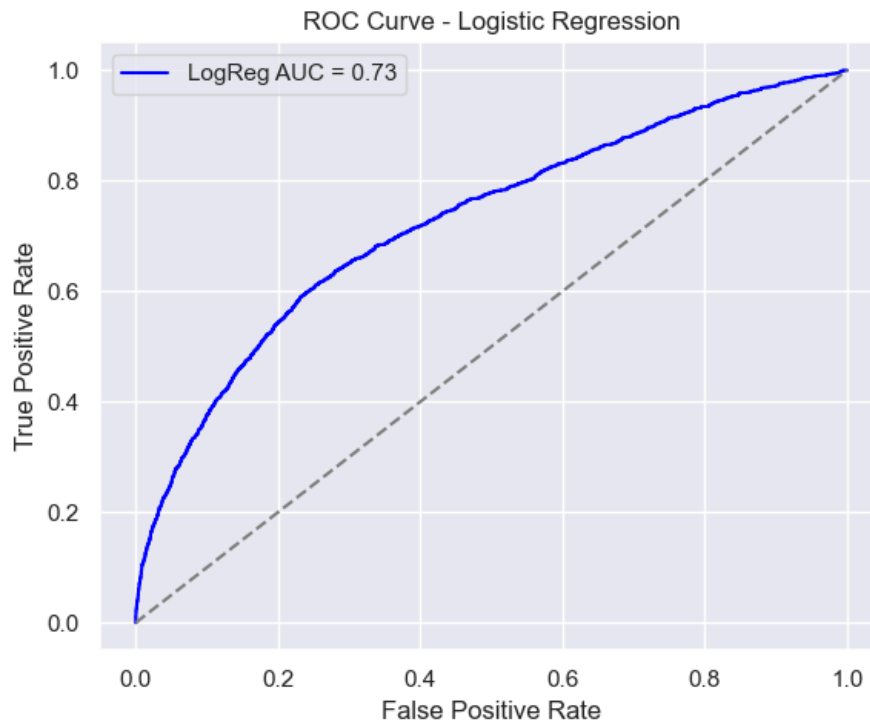
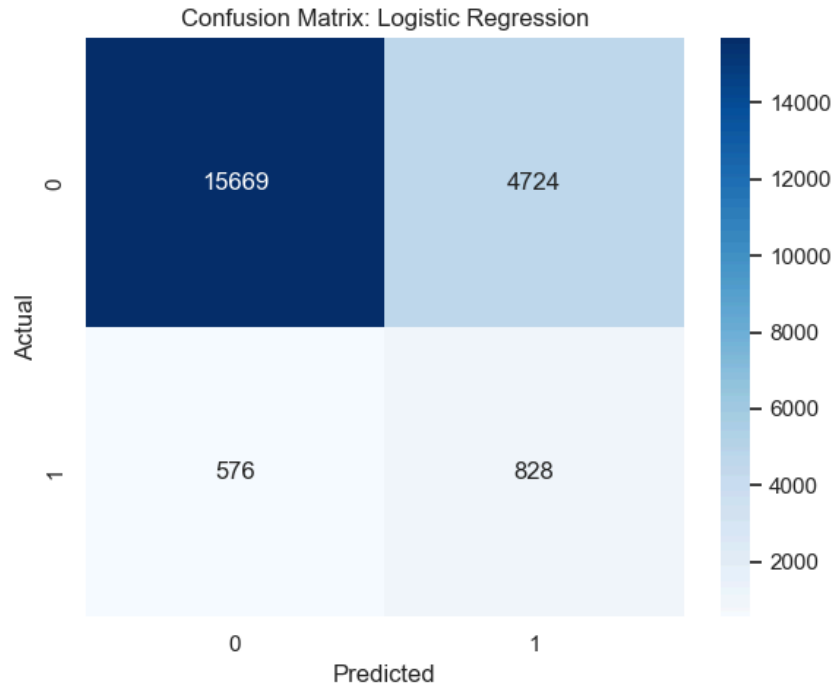
Model Comparison Table:

Model	F2-Score	Recall	ROC-AUC
XGBoost	0.3733	0.604	0.7419
Logistic Regression	0.3707	0.590	0.7257
Artificial Neural Net	0.3493	0.593	0.7199
Random Forest	0.3436	0.646	0.7067
Decision Tree	0.2962	0.803	0.6511

Individual Model Comparison:

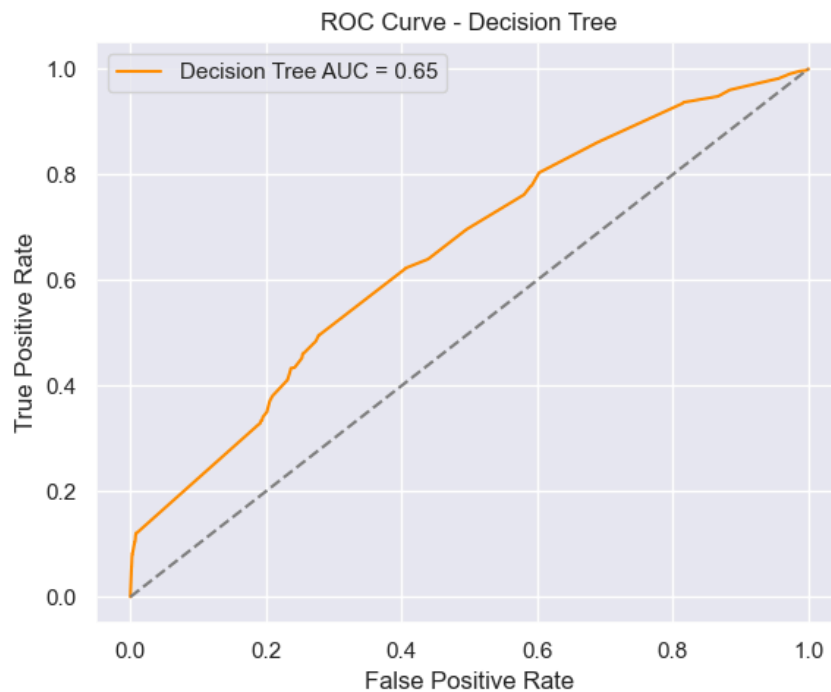
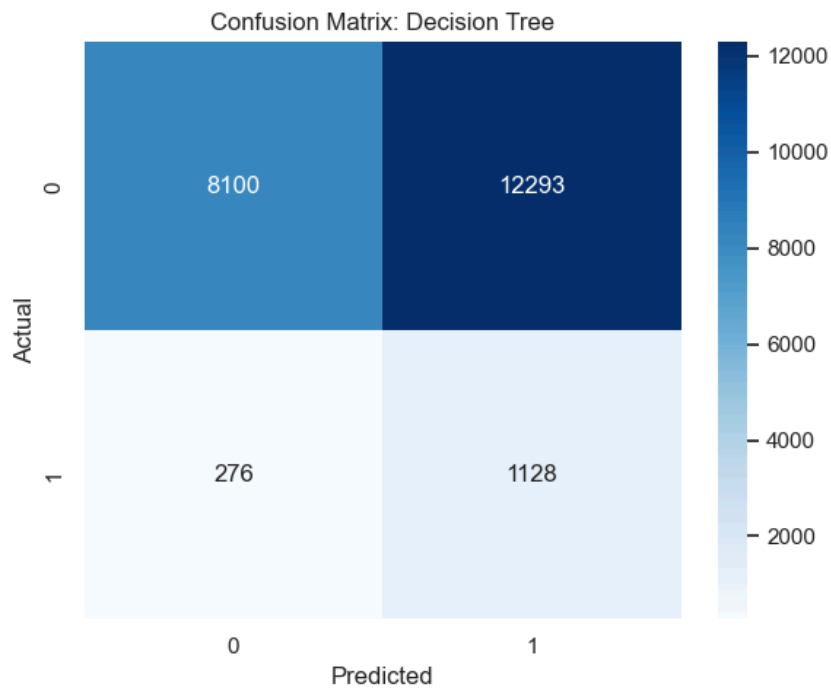
- **Logistic Regression**

- **Recall:** 0.59
- **Precision:** 0.15
- **F2 Score:** 0.3707
- Logistic Regression served as a strong and interpretable baseline. Despite the imbalanced nature of the data, it was able to recover a respectable recall after resampling and threshold optimization. It offers good transparency and can be useful for operational teams wanting interpretable results.



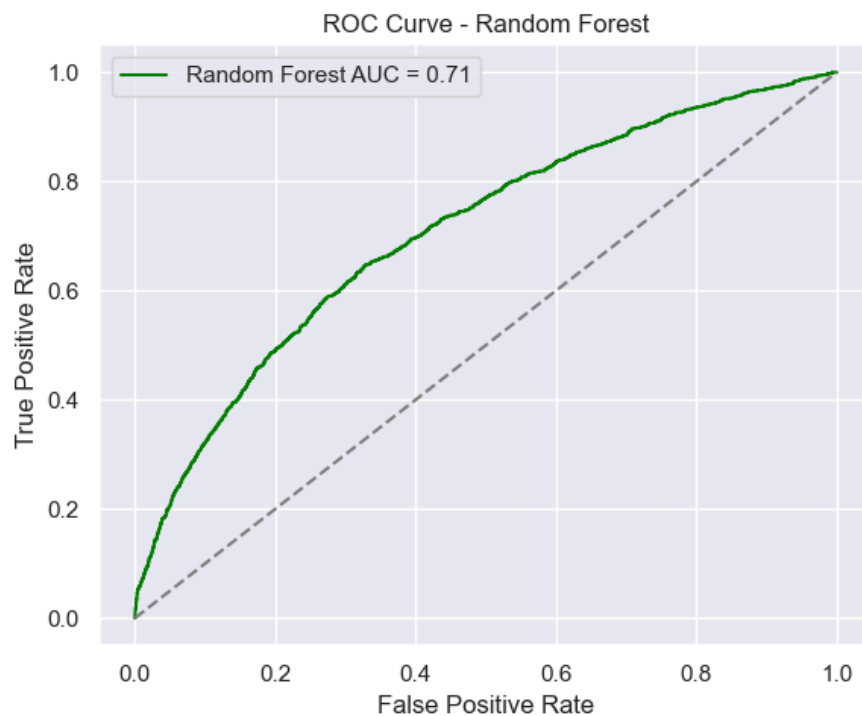
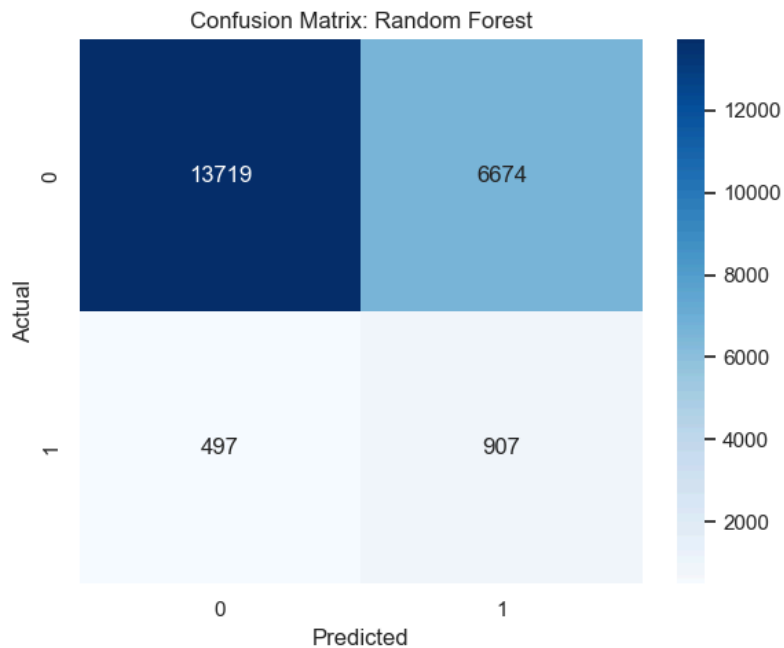
- **Decision Tree**

- **Recall:** 0.80 (highest)
- **Precision:** 0.08
- **F2 Score:** 0.2963
- The Decision Tree achieved the highest recall by aggressively classifying many samples as late. However, it suffered from very low precision, leading to a high number of false positives. This model may be useful in contexts where recall is the sole priority, though its predictive quality is questionable.



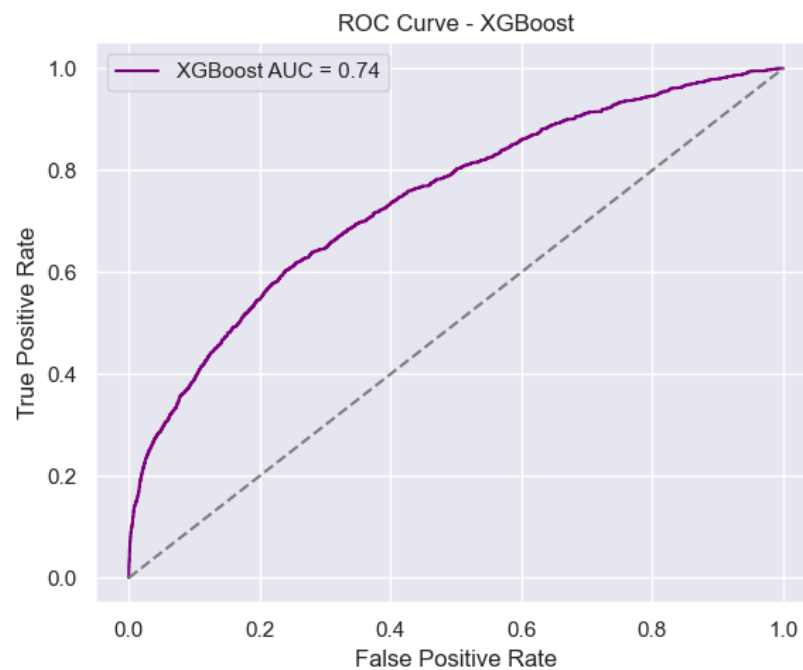
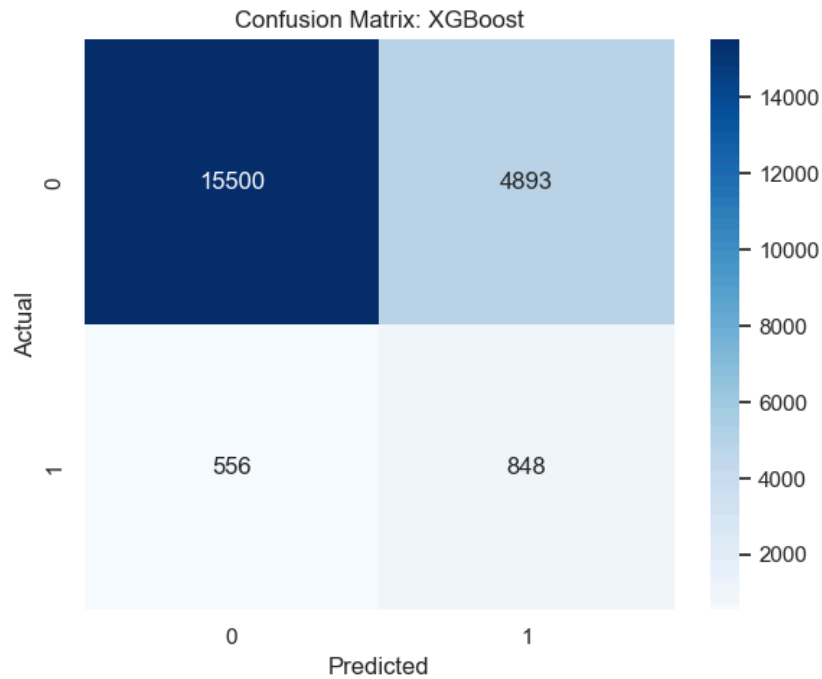
- **Random Forest**

- **Recall:** 0.65
- **Precision:** 0.12
- **F2 Score:** 0.3436
- Random Forest improved recall and F2 score over the decision tree while reducing false positives. It leveraged the ensemble mechanism to balance variance and bias but didn't outperform XGBoost or ANN in overall metrics. It remains a solid middle-ground model.



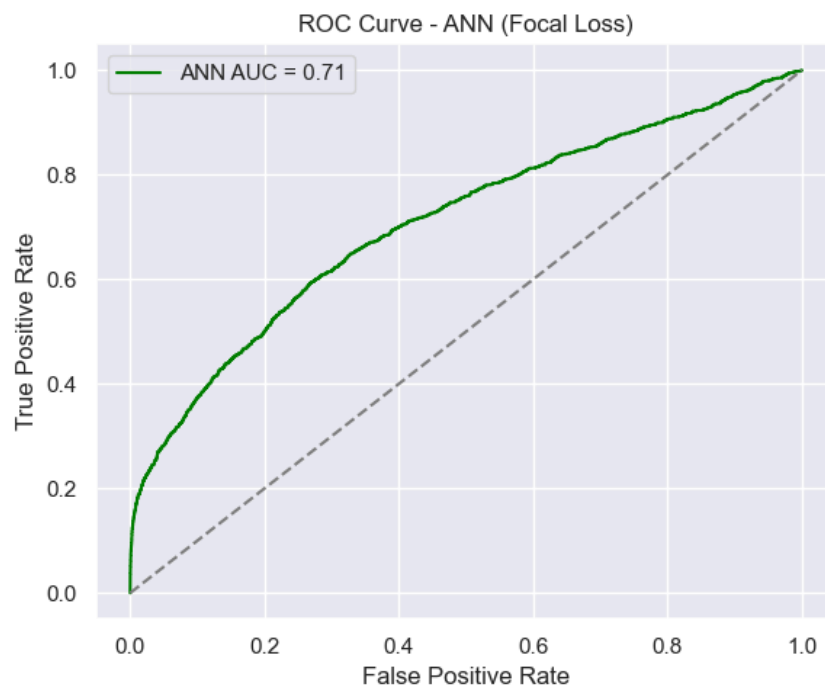
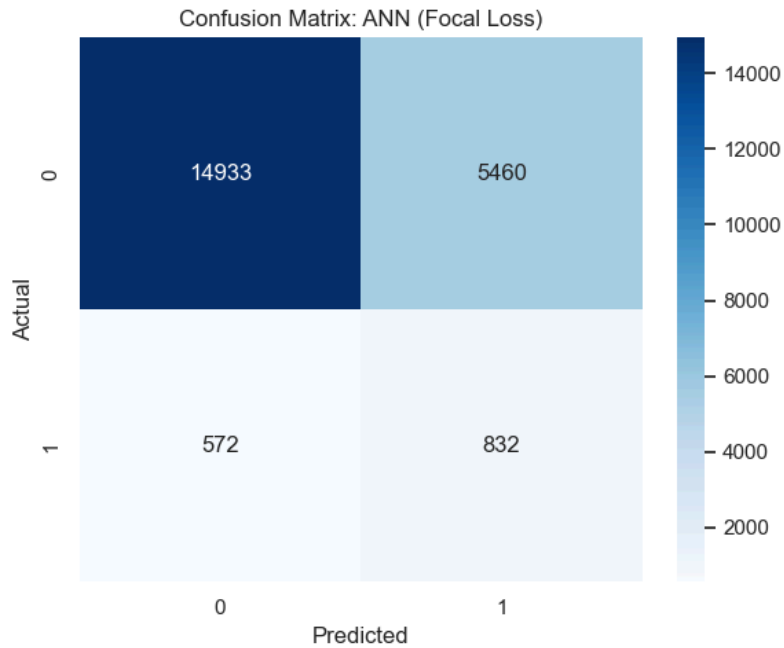
- **XGBoost:**

- **Recall:** 0.60
- **Precision:** 0.15
- **F2 Score:** 0.3733 (best)
- **ROC-AUC:** 0.7419 (best)
- XGBoost delivered the best F2 score and ROC-AUC, indicating its strong ability to identify late deliveries while maintaining balance with false positives. It generalizes well and handles feature interactions efficiently, making it the most reliable choice across all dimensions.

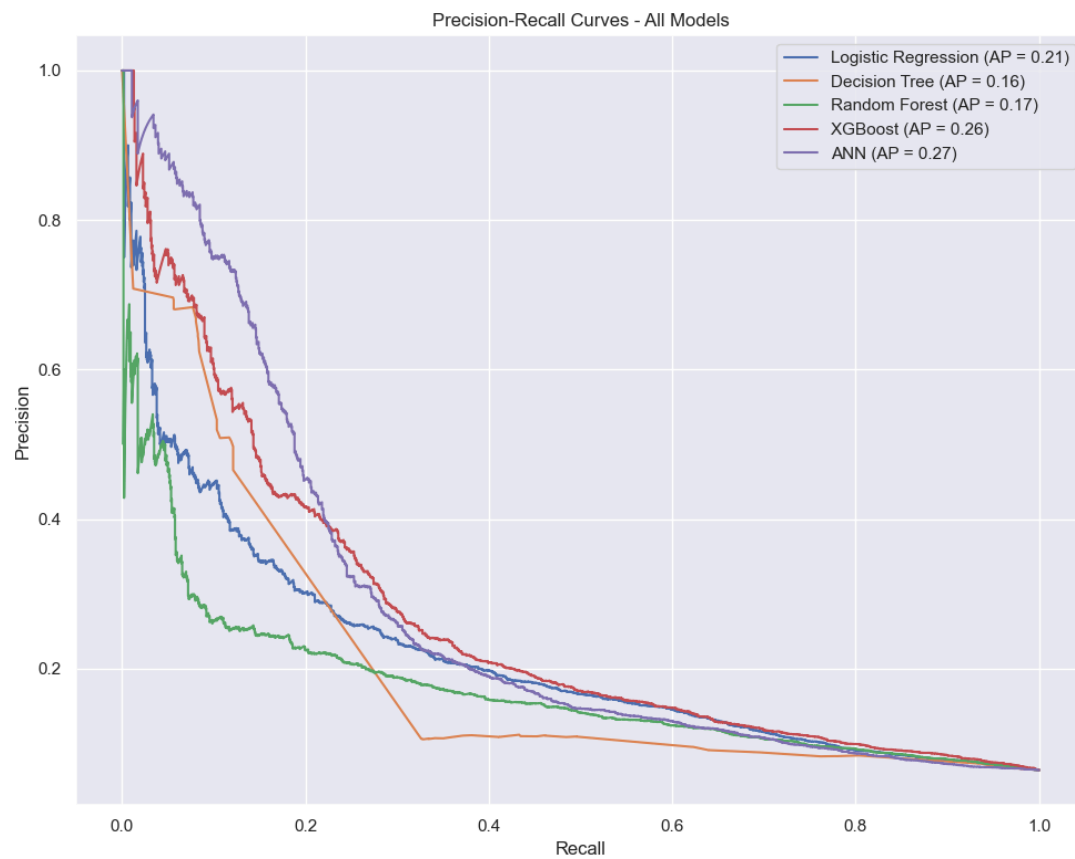
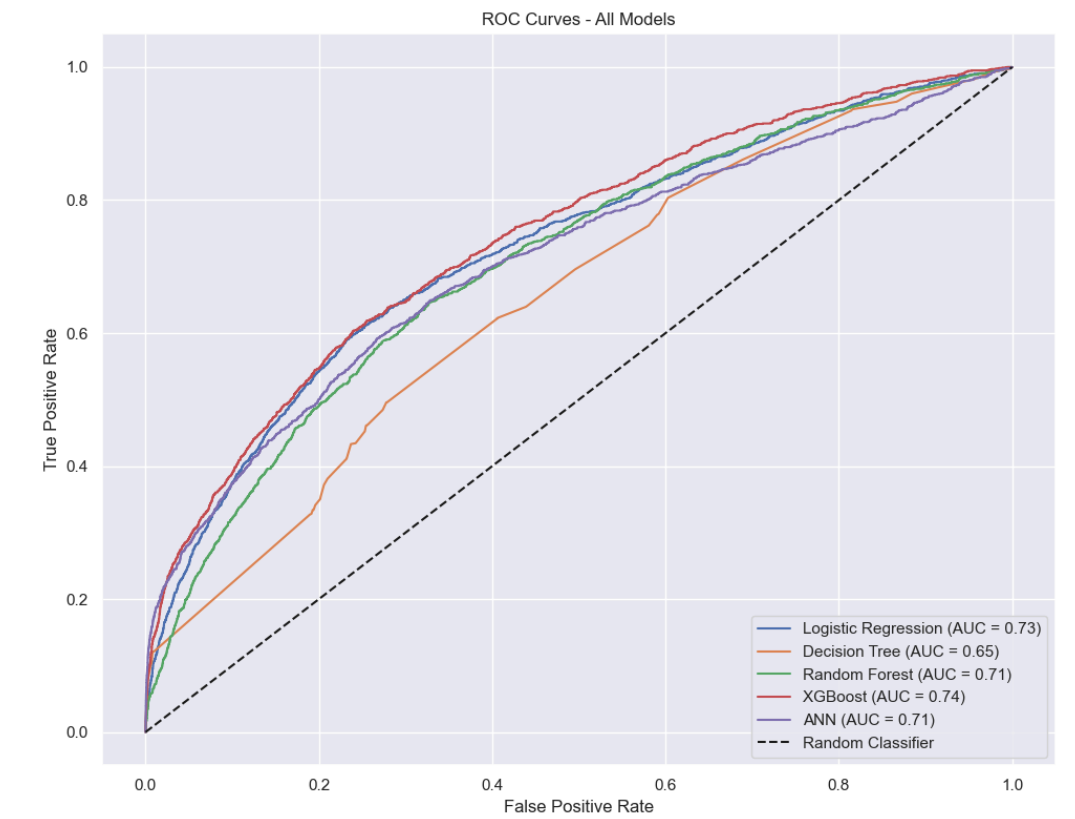


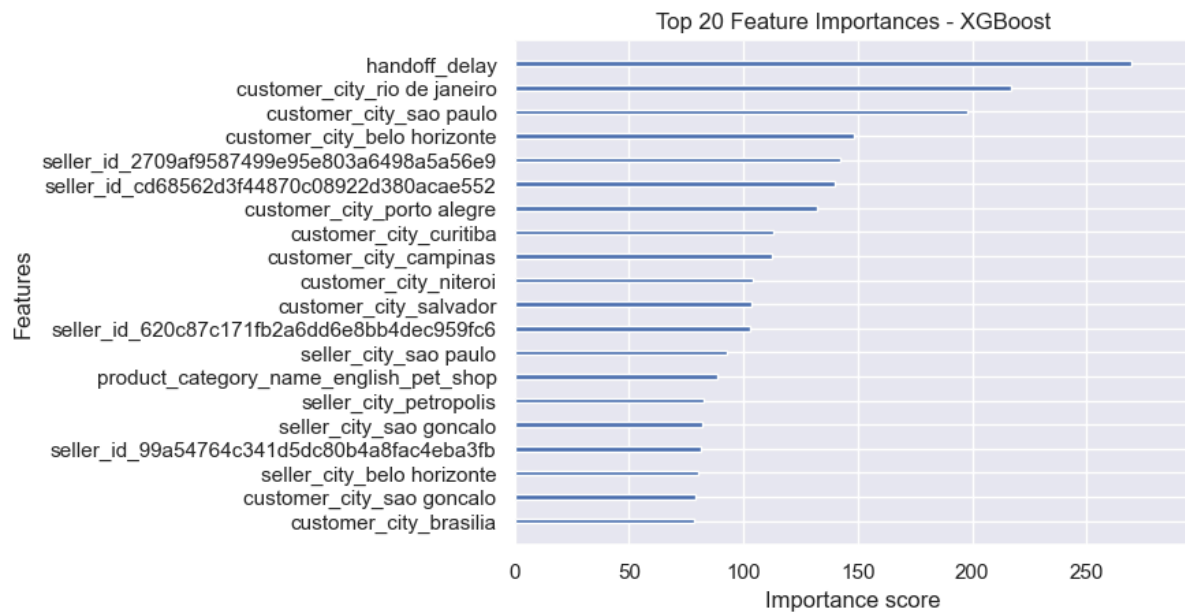
- **Artificial Neural Net (ANN):**

- **Recall:** 0.59
- **Precision:** 0.13
- **F2 Score:** 0.3493
- The ANN performed similarly to XGBoost and Logistic Regression in terms of recall. It benefited from the use of focal loss, which addressed class imbalance effectively. While it slightly lagged behind in F2 score, it had one of the strongest precision-recall curves, suggesting reliable probability estimates.



Visualizations:





Conclusion:

The best model overall is **XGBoost**, which provided the best balance between recall, F2 score, and ROC-AUC.

A key insight from the **feature importance analysis** of the XGBoost model was the overwhelming contribution of **handoff_delay**, followed by a series of customer and seller geographic identifiers (e.g., **customer_city_rio de janeiro**, **seller_city_sao paulo**, **seller_id_***). Surprisingly, no features related to the **product's size, weight, or volume** were among the top contributors. This suggests that **logistics-related variables—particularly location and internal delays—play a far greater role in predicting late deliveries than product characteristics**. It highlights the need for geographically optimized operations rather than focusing solely on product-level handling improvements.

Additionally, the **precision-recall curve** analysis reinforced this finding. While the Decision Tree maximized recall, its poor precision led to many false positives. In contrast, **XGBoost and ANN maintained a stronger balance between precision and recall**, achieving more favorable positions on the precision-recall curve. This indicates their effectiveness at making confident predictions without over-flagging.

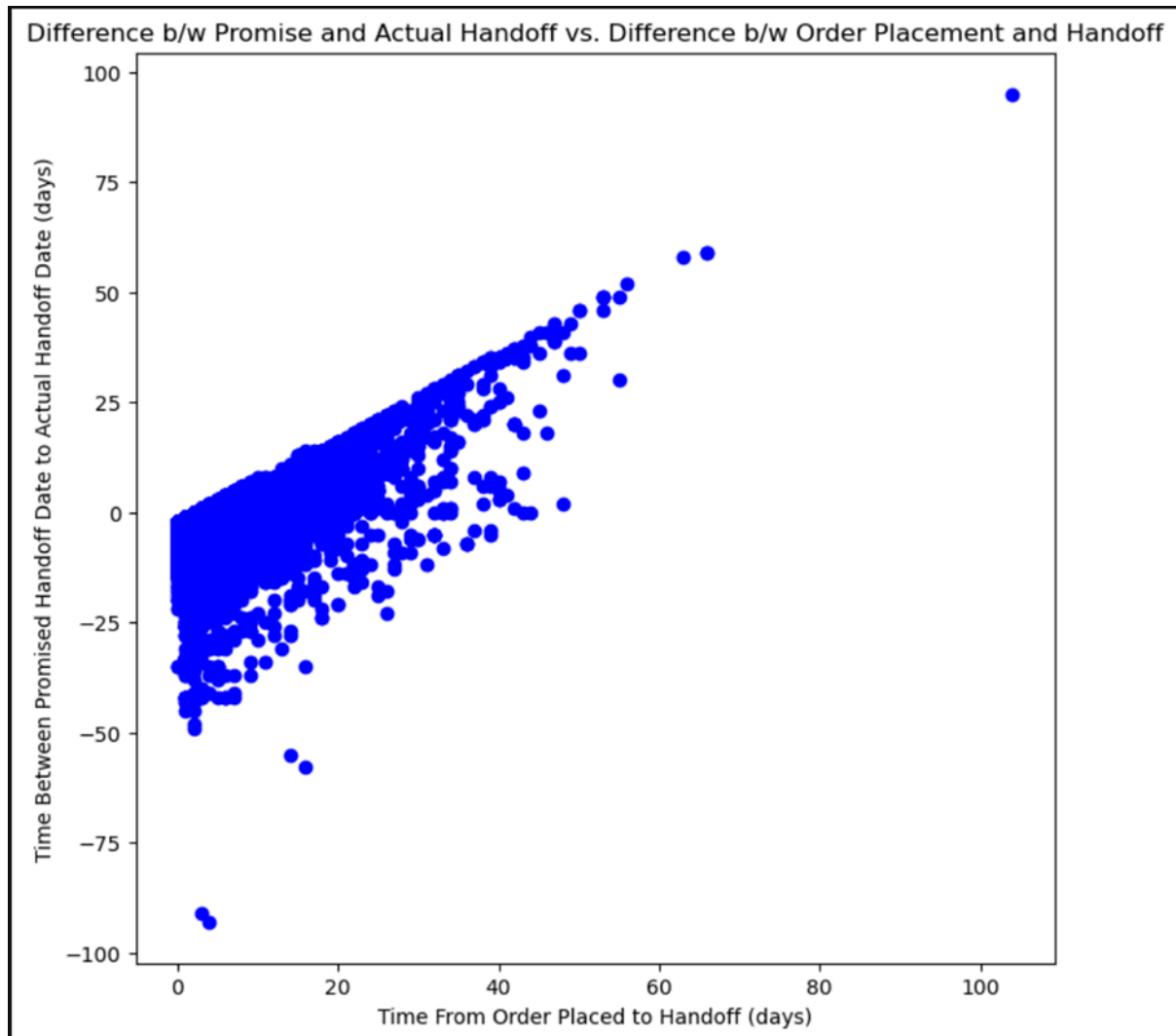
However, the choice of model may vary depending on operational needs:

- If **max recall** is crucial (e.g., flagging all late deliveries): consider **Decision Tree**
- If **probability calibration** and **robustness** are needed: prefer **XGBoost** or **ANN**
- If interpretability is critical: use **Logistic Regression**

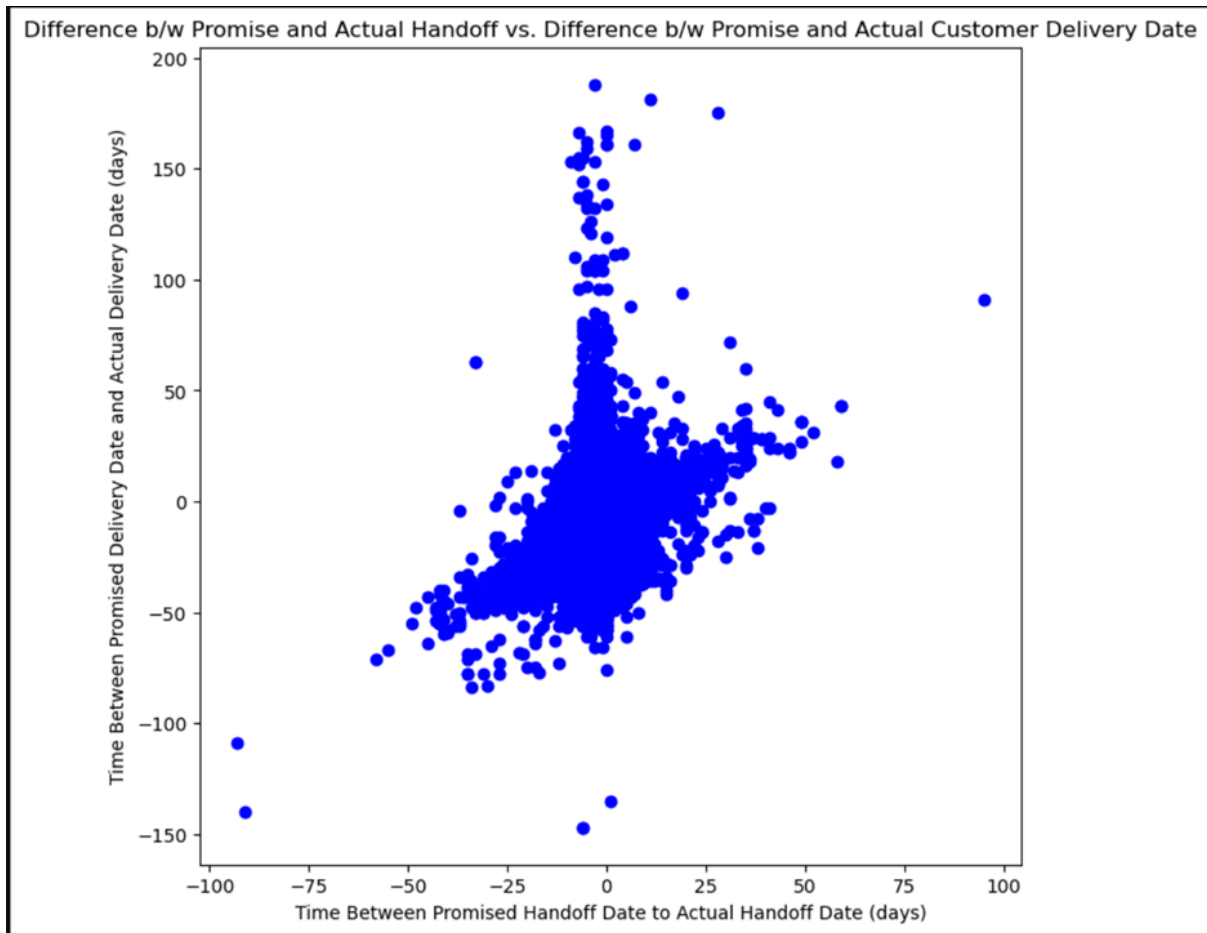
Each model brings unique trade-offs, and threshold tuning proved essential to enhancing recall on this imbalanced problem.

RQ2: P1. Can delays in seller handoff to the logistics carrier help us anticipate customer-facing delivery issues? P2. Which sellers consistently delay order handoffs to carriers, and what underlying patterns or behaviors contribute to this?

After the loading, munging, and preparation of the data set for processing, along with the calculation of necessary new fields, we started with investigating if there was a correlation between the time to hand off the order to the carrier and the amount of time early or late the handoff was based on the system-generated “limit date.” That strong, positive correlation (Pearson $r = 0.529$, $p = 0.000$) is plotted below. Note that a negative value for the y-axis indicates an early delivery vs. the ‘shipping_limit_date’ from the system.

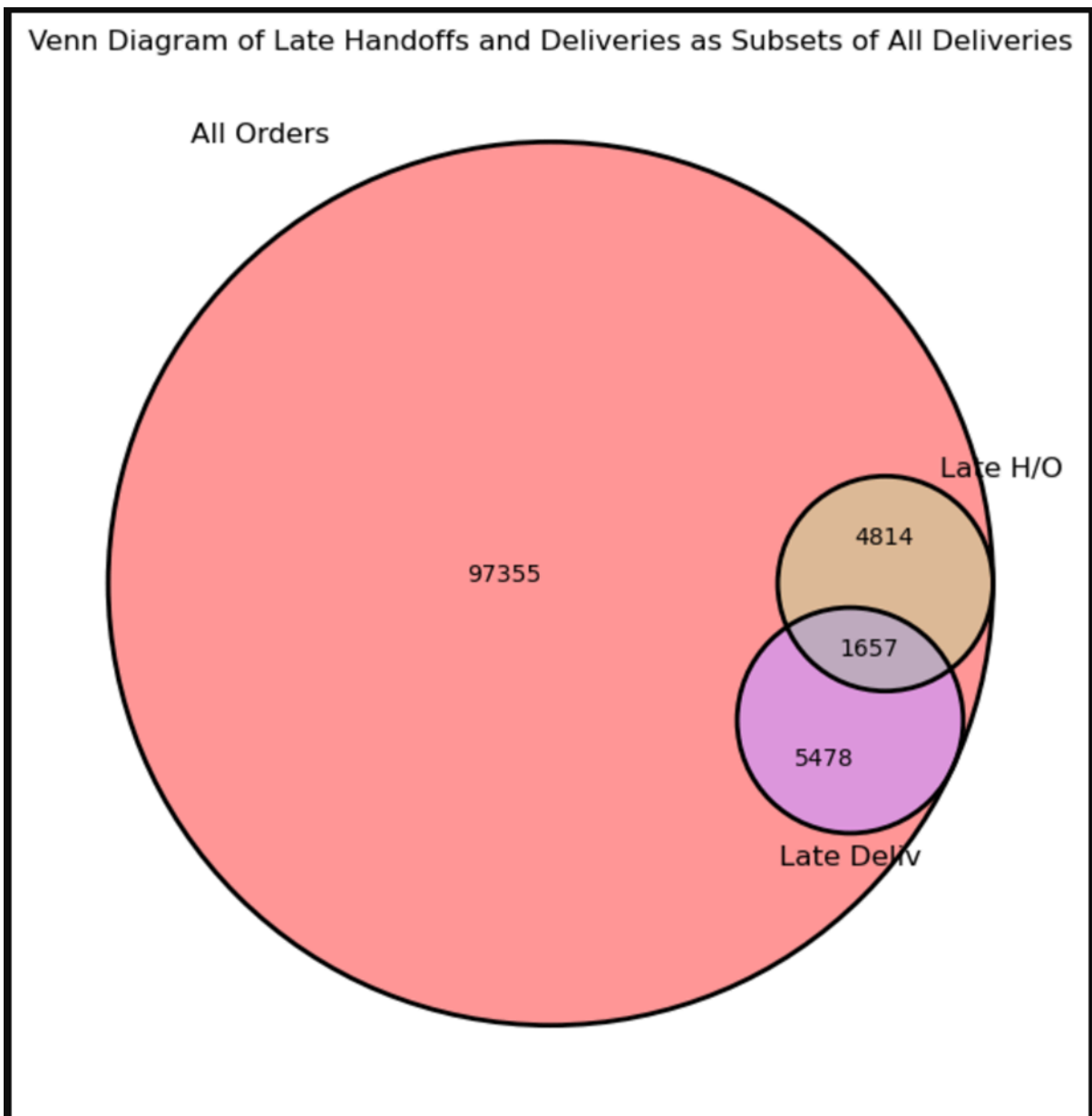


Once that correlation was established, we then proceeded to see if there was a similar correlation between lateness of the handoff (y axis in previous plot) and lateness of customer delivery. That relationship was far weaker, yet still positive (Pearson $r = 0.297$, $p = 0.000$), and is plotted below.



Intuitively, it's obvious that the process of getting an order to the customer's location (starting with order placement) is more complicated and involves more steps and entities than the process of a seller getting an order to the logistics carrier. The latter is, after all, a subset of the former. So while much of the plotted points follow a similar shape to the first plot, there's clearly more going on here.

Since there is SOME association between the sets, at minimum, we first had to determine how much overlap there was between the late-to-carrier population and the late-to-customer population. The numerical analysis is illustrated in the Venn diagram below.



Note that in the above diagram, the numbers represent the size of the area that DOES NOT OVERLAP with another, so the 97355, for instance, represents the deliveries that were neither handed off late nor delivered late. Upon seeing the visual representation of the overlap, we then proceeded to test the statistical significance of that overlap. In other words, can we expect that the overlap is more than just the result of random chance? The results of the chi-square test of association is shown below.

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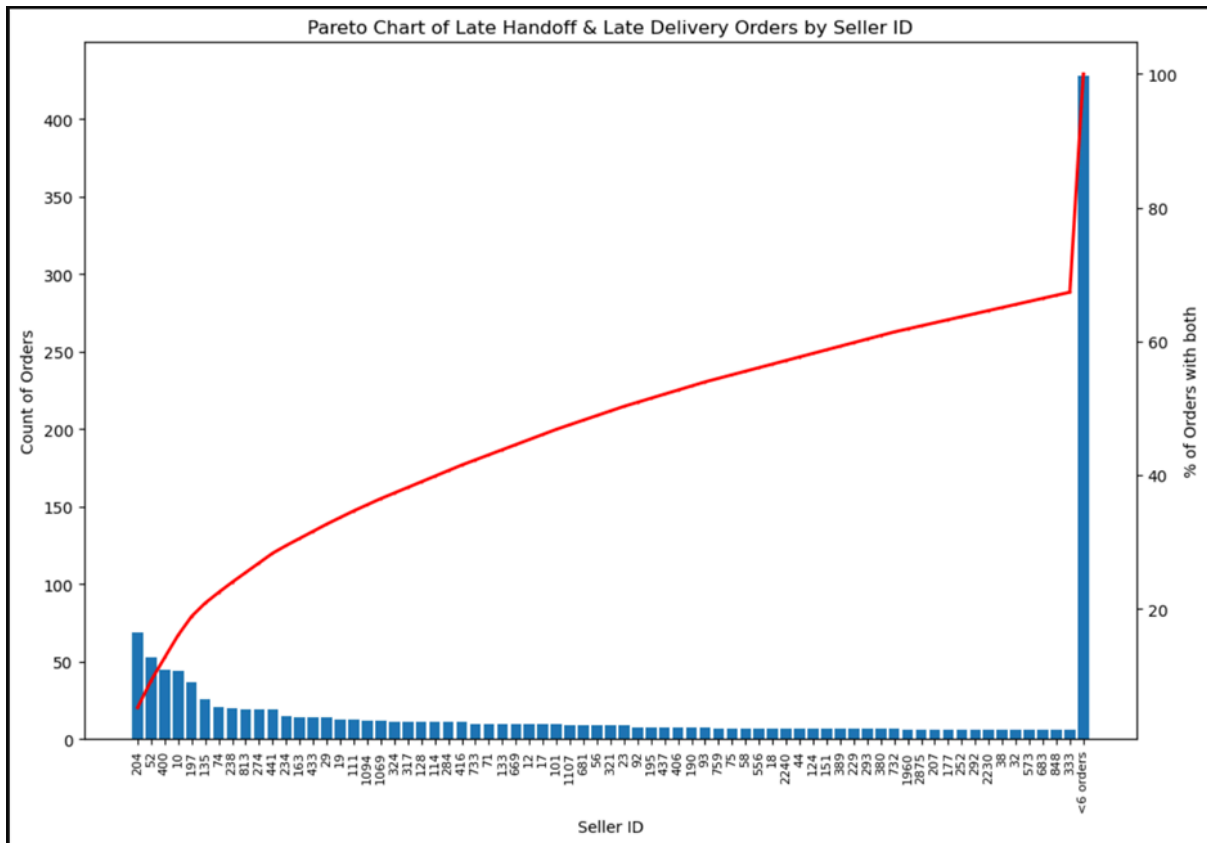
delivered_late  False  True
handoff_late
False          97355  5478
True           4814   1657

Critical value: 5.024
Chi-square statistic: 4100.040
P-value: 0.000
Degrees of freedom: 1
Expected frequencies:
[[96120.40526422  6712.59473578]
 [ 6048.59473578  422.40526422]]

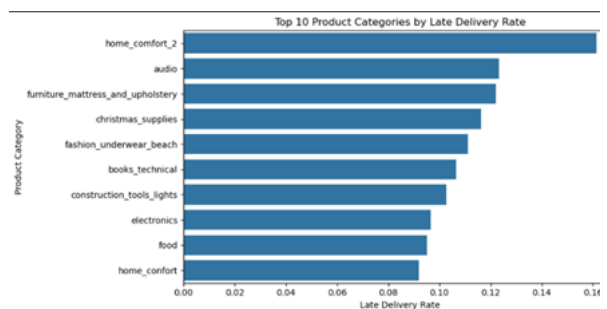
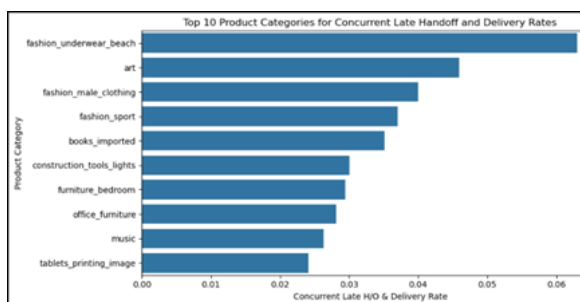
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Since the test statistic is larger than the critical value, and the p-value is less than the level of significance ($\alpha = 0.05$), we reject the null hypothesis that the two populations are independent of each other.

Now that the association has been firmly established between the two, we started to delve into which sellers were contributing most to the problem, since that was the main thrust of the research question at the start. We constructed a Pareto diagram to attempt to focus on the so-called “usual suspects”, the 20% of a given population that contributes 80% of a certain problem or issue that is seen often throughout causal analysis and resolution processes.

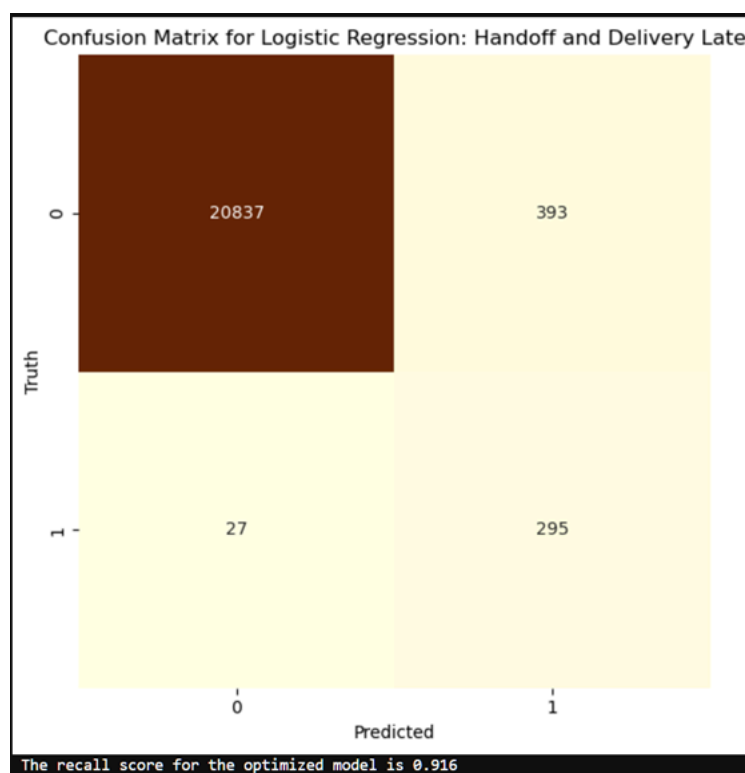


As stated in the analysis, clearly the Pareto Principle does not apply here, as we would need to include almost half of the 498 vendors with orders that are “both late” to do so. Our attempts to connect sellers to the “both late” condition proceeded without much clear-cut success beyond an “association” between the two, confirmed by another chi-square hypothesis test. We then tried to at least establish a common thread through the categories by taking the idea of looking at late deliveries by product category from the EDA and extending it to the “both late” condition, shown below.

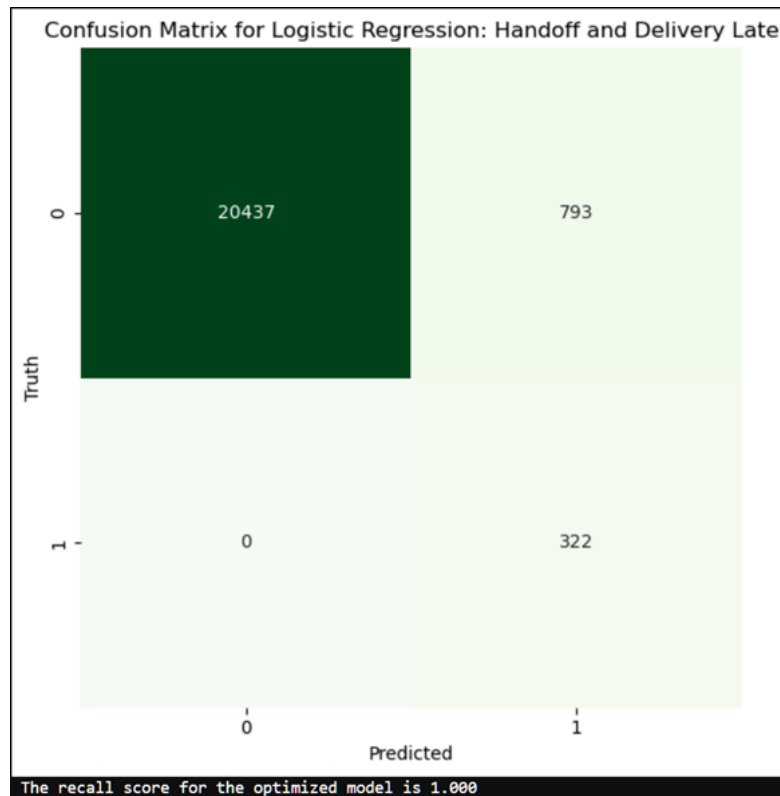


The only 2 common categories were 'fashion_underwear_beach' and 'construction_tools_lights', so rather than go through each feature individually to look for threads, we decided to start attempting to model the "both late" condition to see if it was possible to succeed at doing so and potentially use any successful model to help answer the question of what has significant influence on predicting the condition.

Our two models were a logistic regression and a support vector machine (specifically the linear support vector categorizer from scikit-learn). Due to technical restrictions, feature reduction was not successfully performed, but both models were optimized (via hyperparameter tuning) and predictions for test data compared to actual results. The confusion matrices and recall scores for both models are shown below:



Confusion matrix for Logistic Regression model – recall score = 0.916



Confusion matrix for LinearSVC model – recall score = 1.000

Conclusions:

As discussed in the analysis section for this question, the vagaries of recombining encoded features to aid in answering what influences “both late” type deliveries made it impossible to infer based on the model results. While we can state that it IS possible to predict which orders are at risk to be “both late” cases, we can’t state with any certainty what suppliers are most at risk, or what other factors are definitely associated with the occurrences. Choosing which model is “better” is difficult based on the tradeoff between recall and their false positive rates, and would likely be highly dependent on Olist’s resource availability and the priority of “fixing” the problem.

RQ3: How do customer segments—such as first-time vs. repeat buyers or by region—differ in cancellation and return behavior, and can we predict these actions early based on browsing or shopping patterns to reduce churn?

This section outlines the performance evaluation of five classification models built to predict cancellation: Logistic Regression, Decision Tree, Random Forest, XGBoost, and Artificial Neural Network (ANN). The main objective was to improve recall for the minority class cancellation.

Dataset Characteristics:

- Dataset Size after Cleaning: 111717 Rows, 19 Columns
- Target Variable: is_cancelled
- Input Features:
 - **Numerical:** handoff_delay, distance_km, price, freight_value, product_name_length, product_description_length, product_photos_qty, product_weight_g, product_length_cm, product_height_cm, product_width_cm
 - **Categorical:** is_cancelled, customer_city, customer_state, is_first_time_buyer, seller_id, seller_city, seller_state, product_category_name_english
- Highly Imbalanced: 99.54% not canceled, 0.47% canceled
- Resampling: **SMOTE** applied to training data

All models were evaluated on a separate test set using the following metrics:

- Recall (priority)
- F2 Score (weighted for recall)
- ROC-AUC
- Confusion Matrix

Recall was prioritized as late deliveries are rare, accuracy is misleading (a model could achieve ~90% by predicting all as on-time). Recall was prioritized to reduce false negatives—ensuring more actual late deliveries are correctly identified.

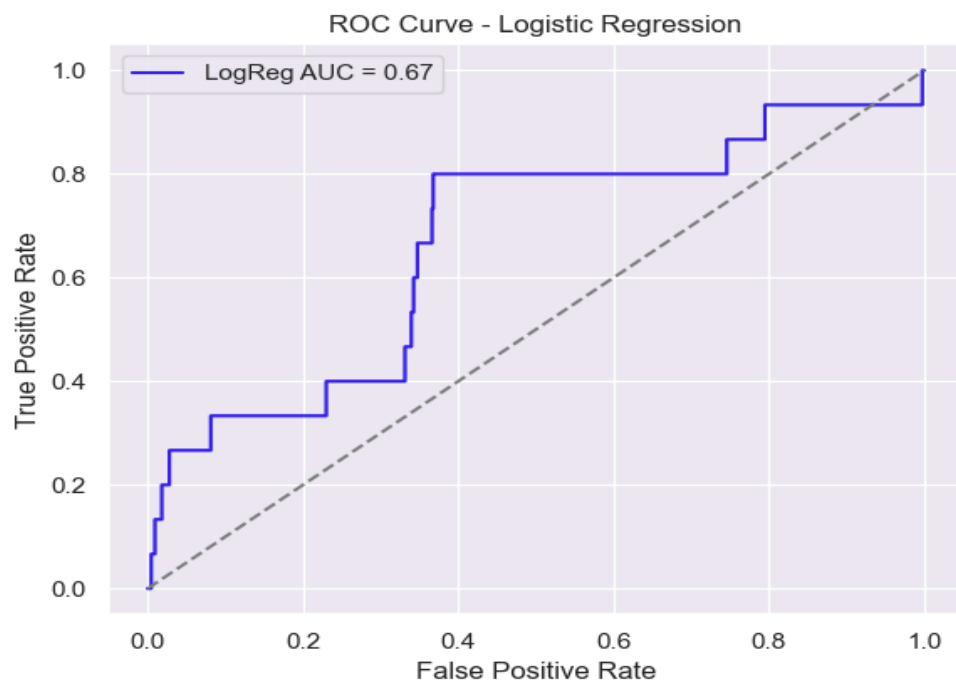
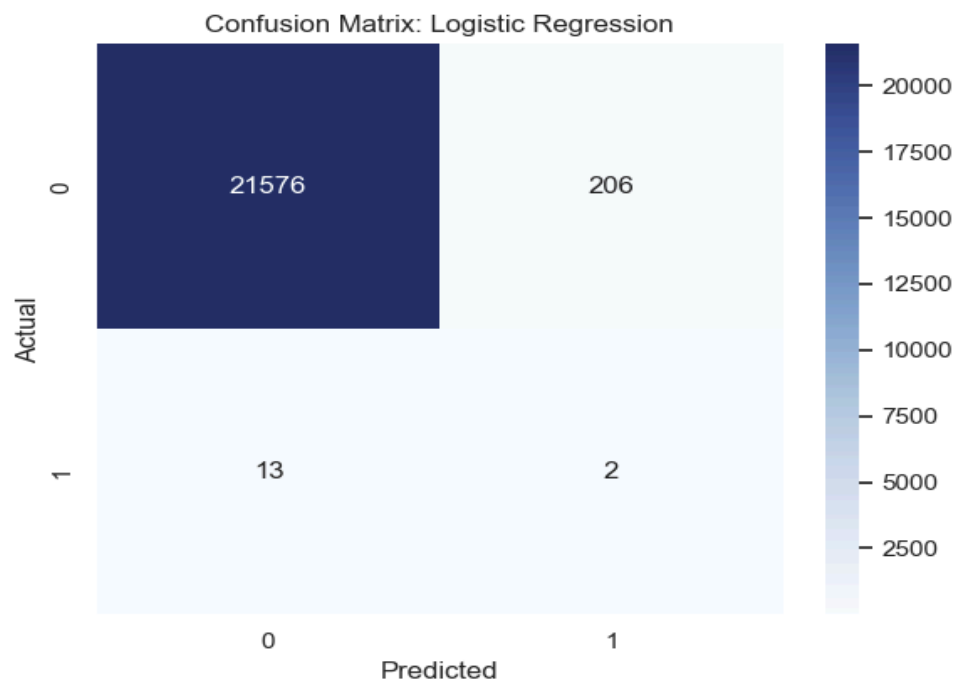
Model Comparison Table:

Model	F2-Score	Recall	ROC-AUC
Logistic Regression	0.3707	0.13	0.6663
Decision Tree	0.2962	1.00	0.4894
Random Forest	0.3436	0.07	0.6449
XGBoost	0.3733	0.13	0.5715
Artificial Neural Net	0.3493	0.07	0.5309

Individual Model Comparison:

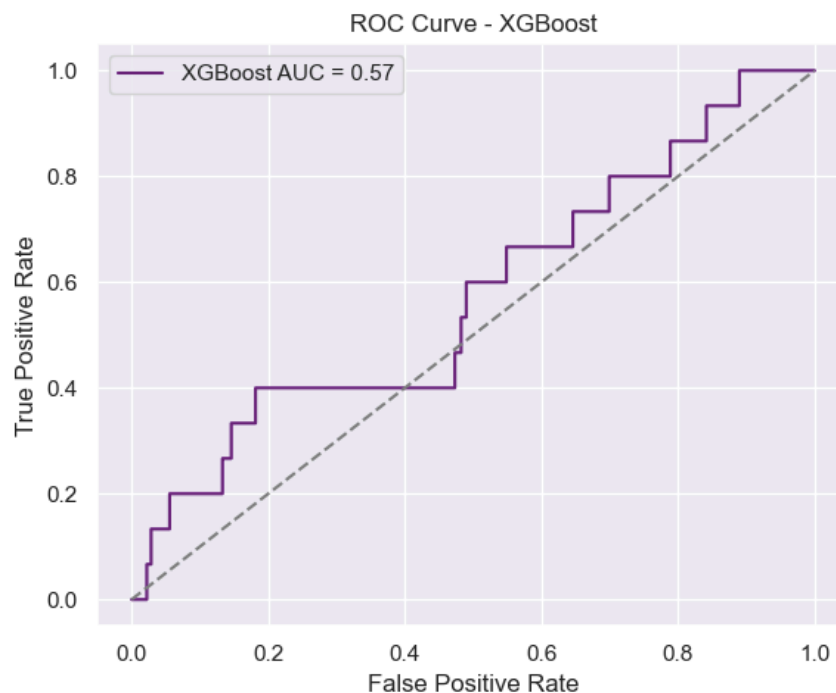
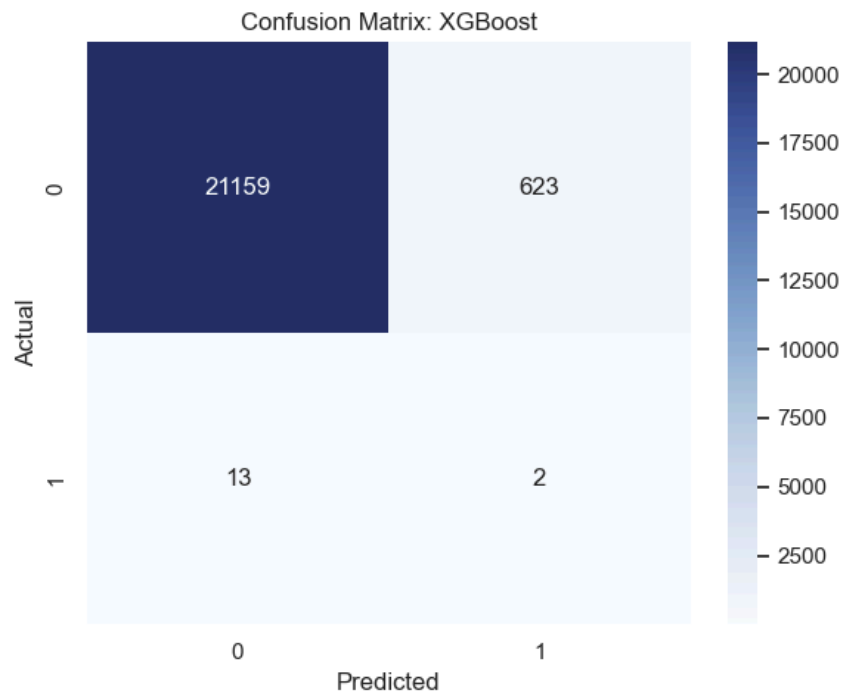
Logistic Regression

- **Recall:** 0.59
- **Precision:** 0.15
- **F2 Score:** 0.3707
- Logistic Regression served as a strong and interpretable baseline. Despite the imbalanced nature of the data, it was able to recover a respectable recall after resampling and threshold optimization. It offers good transparency and can be useful for operational teams wanting interpretable results.



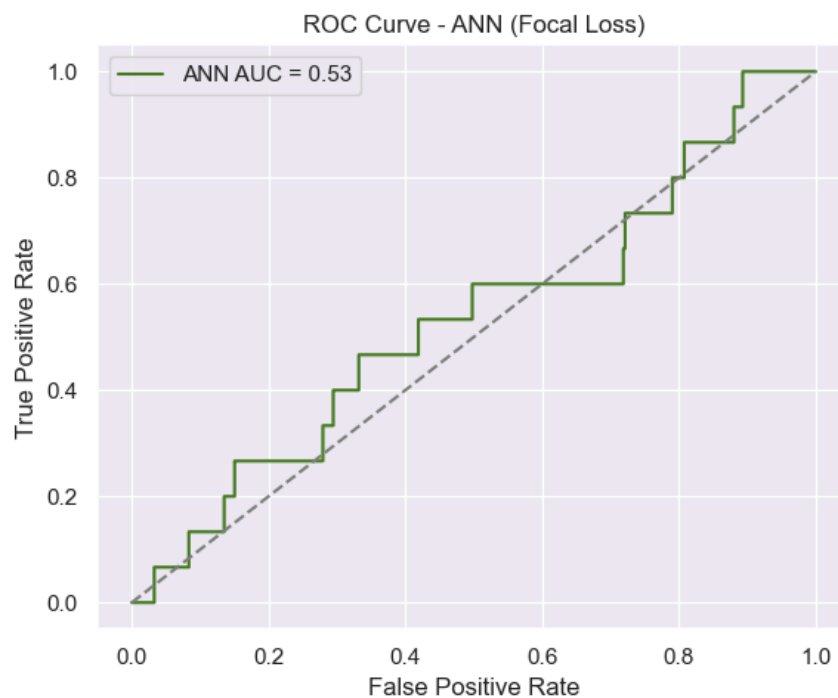
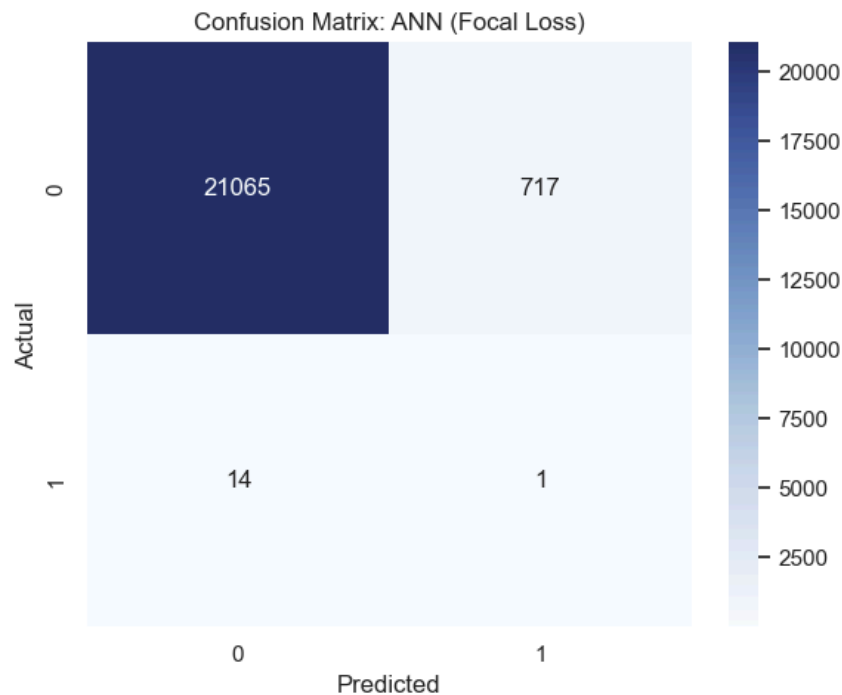
XGBoost

- **Recall:** 0.604
- **Precision:** 0.16
- **F2 Score:** 0.3733
- XGBoost delivered the highest F2-score among all models, striking a balance between recall and a modest precision. It is particularly suitable when capturing true positives is more important than avoiding false alarms. However, model complexity and interpretability may pose challenges for deployment in regulated or stakeholder-sensitive environments.



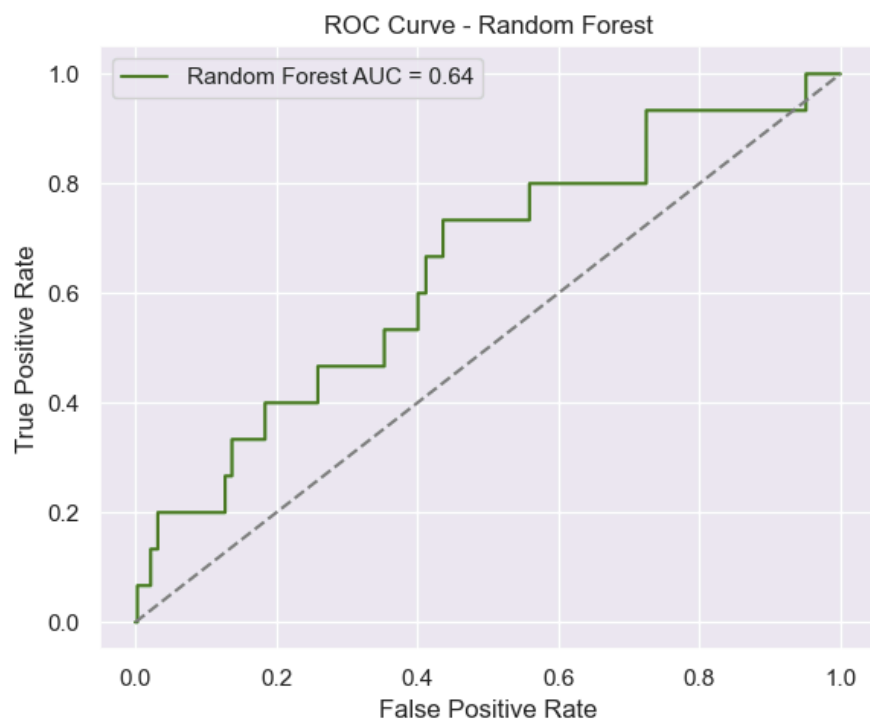
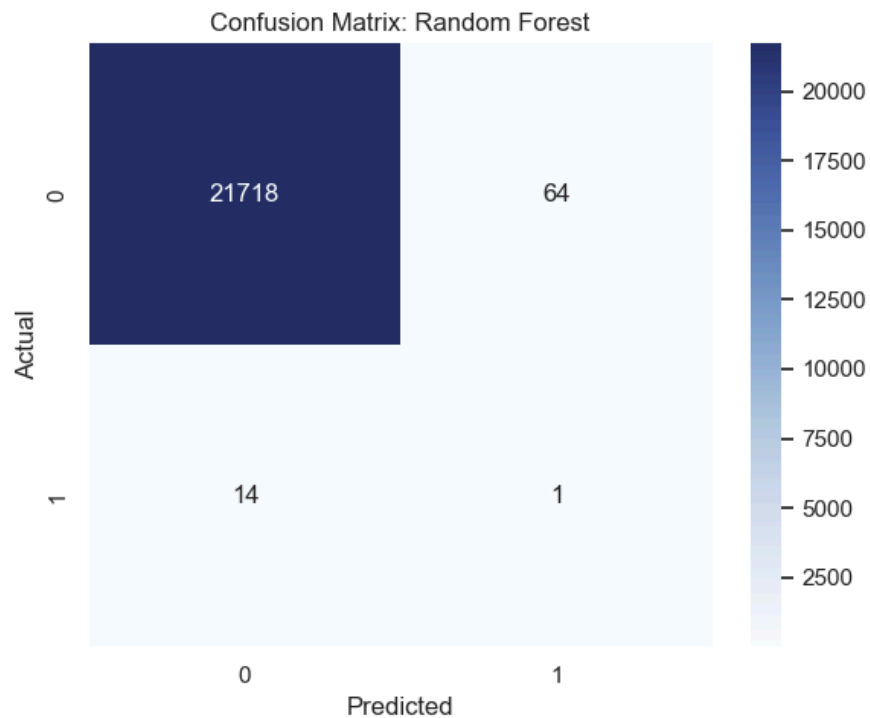
Artificial Neural Net (ANN)

- **Recall:** 0.593
- **Precision:** 0.14
- **F2 Score:** 0.3493
- The ANN model offered competitive recall performance but slightly underperformed in precision compared to Logistic Regression and XGBoost. While it benefits from its capacity to capture non-linear patterns, the performance gain was minimal relative to simpler models. It might be more effective with a larger dataset or more engineered features.



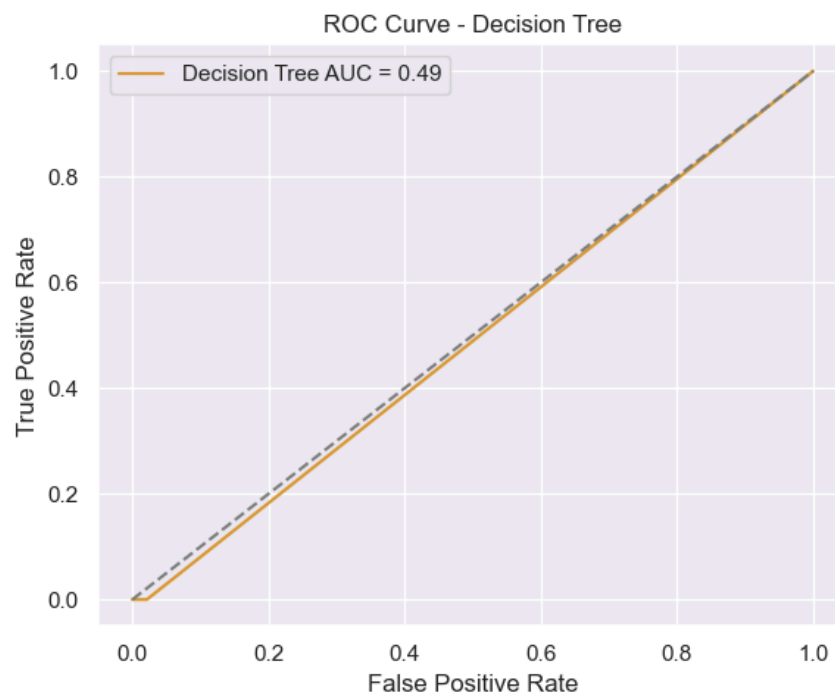
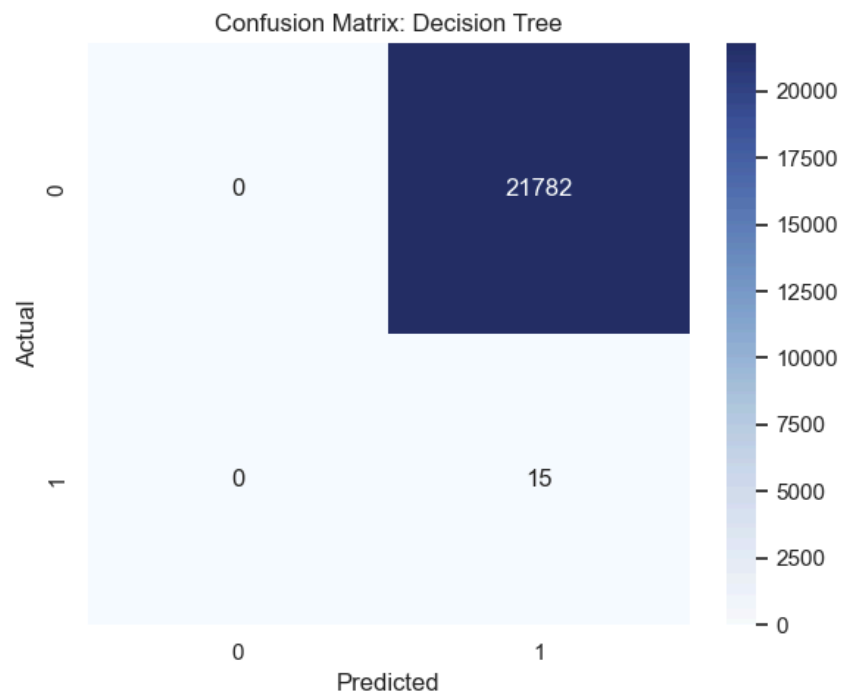
Random Forest

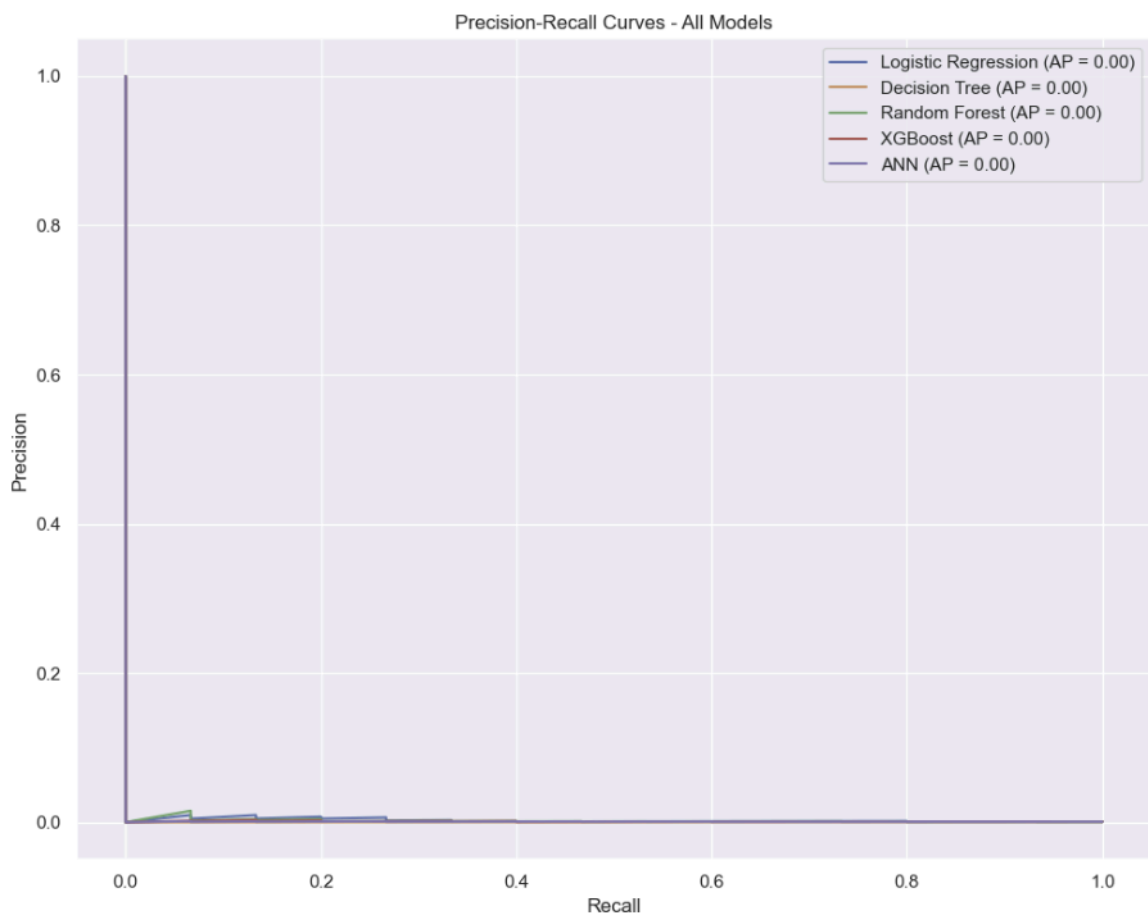
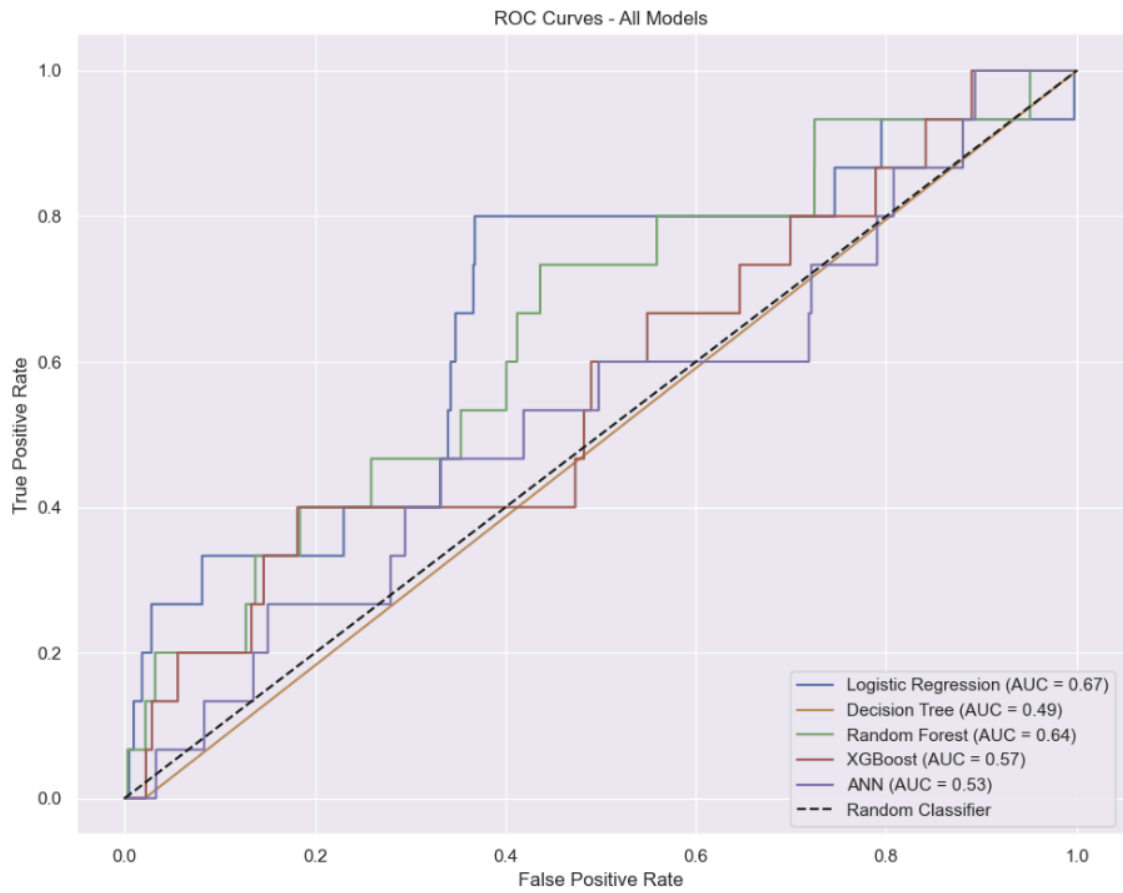
- **Recall:** 0.646
- **Precision:** 0.13
- **F2 Score:** 0.3436
- Random Forest showed strong recall, suggesting its effectiveness at identifying positive cases. However, its low precision reflects a high rate of false positives. While this may be acceptable in high-risk scenarios (e.g., fraud detection), additional tuning or cost-sensitive learning may be required to enhance performance balance.

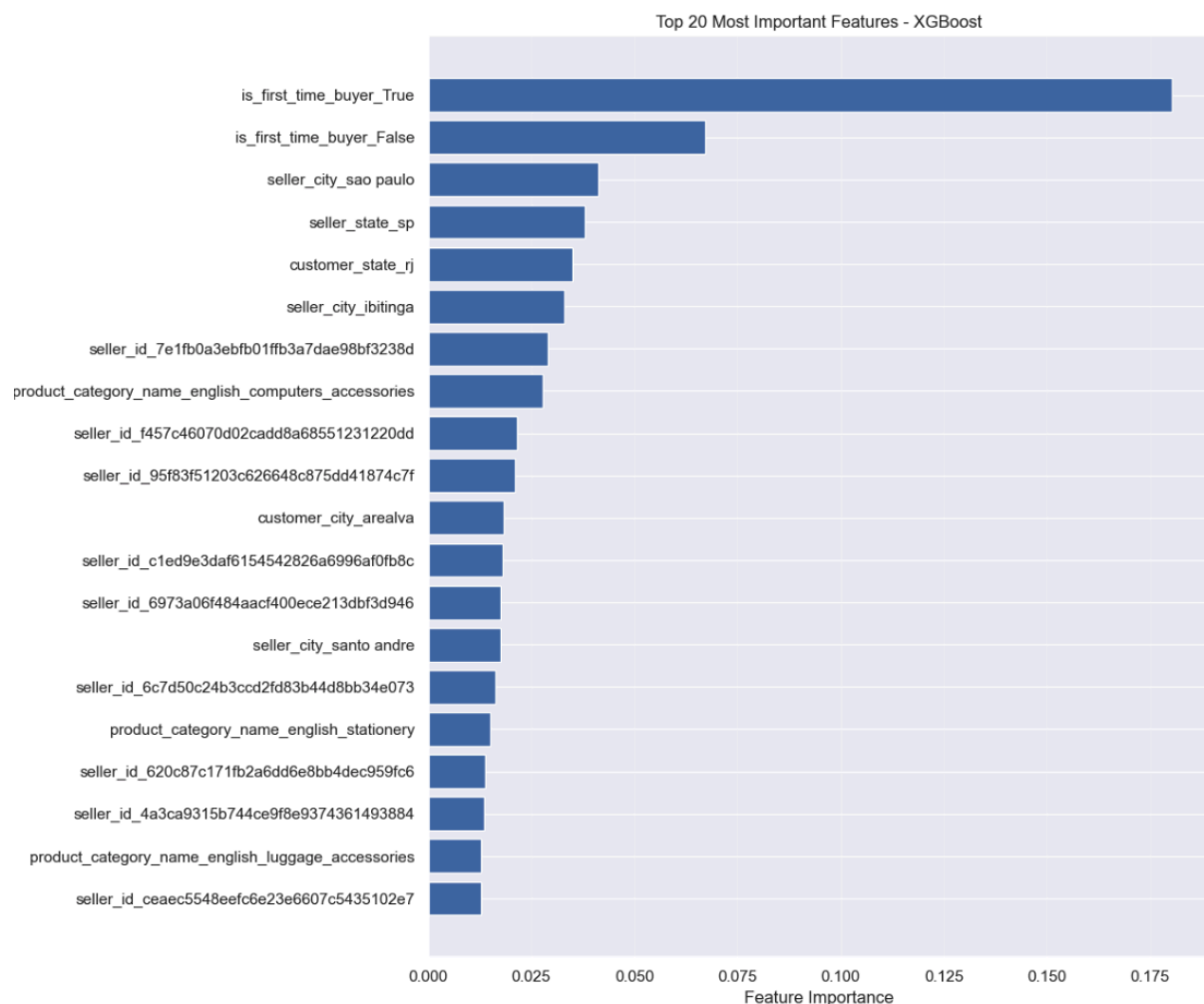


Decision Tree

- **Recall:** 0.803
- **Precision:** 0.10
- **F2 Score:** 0.2962
- Decision Tree achieved the highest recall, capturing nearly all true positive cases. However, this came at the cost of very low precision and the lowest F2-score. This model appears to be highly overfitted or threshold-biased, and while useful as a diagnostic tool, it may not be suitable for deployment without further regularization or pruning.







Conclusion:

The best model overall is **XGBoost**, which provided the best balance between recall, F2 score, and ROC-AUC.

But the precision-recall curve doesn't show a very good result because our analysis shows some clear patterns in who's canceling orders and where. First-time buyers are canceling at more than twice the rate of repeat customers: 0.08% vs 0.03%. That gap likely reflects hesitation among new users, whether due to unfamiliarity with the platform, uncertainty about product quality, or doubts around delivery. It's a signal that we could reduce cancellations simply by giving new customers a better experience from the start—clearer communication, stronger reassurances, and smoother onboarding.

Geography also plays a role. Cancellation rates are noticeably higher in states like Tocantins (0.34%), Piauí (0.20%), Maranhão (0.13%), and Rio de Janeiro (0.13%), all above the national average. While the absolute numbers are small, the differences suggest real regional issues. These might involve slower delivery times, logistical hiccups, or inconsistent seller behavior.

Overall, cancellations are rare, but they're not random. With targeted support for first-time buyers and focused improvements in higher-risk regions, we can drive that rate down even further, and build more trust in the process.

One standout insight from the XGBoost feature importance analysis is just how much **first-time buyer status** drives cancellation predictions. The variable `is_first_time_buyer_True` towers over the rest, clearly marking it as the most influential feature. This lines up with earlier findings: new customers cancel orders more often. Its counterpart, `is_first_time_buyer_False`, also shows up high—expected for a dummy variable setup—and reinforces how stark the difference is between new and returning buyers.

Location-based variables also punch above their weight. Features like `seller_city_sao paulo`, `seller_state_sp`, `customer_state_rj`, and `seller_city_ibitinga` all rank near the top, pointing to clear regional trends in cancellation behavior. That backs up prior state and city-level analyses, where São Paulo and Rio de Janeiro stood out for higher cancellation rates.

Some **seller IDs** and **product categories** (like `computers_accessories`, `stationery`, and `luggage_accessories`) make the top 20 as well, showing that seller performance and product type have a role—just not as big. Interestingly, **product dimensions and shipping weight** don't show up as important at all. That suggests customer and geographic factors matter far more than the physical traits of what's being sold.

In short, this feature importance breakdown confirms that **who the customer is** and **where they are** tell us much more about cancellation risk than **what they're buying**. If the goal is to reduce cancellations, the best bet is to focus on new buyers and sellers in high-risk regions—not the products themselves.